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Advances in dicarbofunctionalization of alkenes, alkynes and allenes *via* iron catalysis

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Transition metal-catalyzed dicarbofunctionalization of unsaturated bonds, which enables the simultaneous installation of two carbon substituents at vicinal positions, offers an efficient strategy for constructing complex molecular architectures. This approach is particularly valuable in the synthesis of natural products and pharmaceuticals, where rapid diversification of carbon frameworks is essential. While transition metals such as palladium have traditionally dominated this area, concerns over their limited availability and high cost have driven the development of more sustainable catalytic systems. In recent years, iron-catalyzed dicarbofunctionalization of unsaturated bonds has emerged as a powerful alternative for the formation of diverse C–C bonds. This review provides a systematic overview of iron-catalyzed three-component dicarbofunctionalization of alkenes, alkynes, and allenes employing organomagnesium, organozinc and organolithium reagents as the coupling partners. Particular emphasis is placed on the practical applications of these methods, along with detailed discussions of reaction mechanisms and selectivity control. Finally, current limitations are discussed, and future directions are proposed, including strategies for achieving stereocontrol and enabling reductive couplings between two electrophilic partners.

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1. Introduction

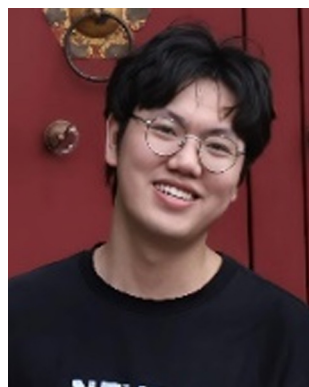
Over the past few decades, transition metal-catalyzed cross-coupling reactions have become indispensable tools for organic

chemists to construct carbon–carbon and carbon–heteroatom bonds. Traditionally, noble metals such as palladium and rhodium have played crucial roles in this field. However, concerns over their scarcity and high cost have spurred the development of more sustainable catalytic approaches. In this context, iron catalysis has emerged as a promising alternative, offering distinct advantages such as low cost, high natural abundance, low toxicity, and environmental compatibility.¹ Notably, iron-catalyzed cross-coupling reactions have already been successfully implemented in kilogram-scale industrial

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processes, demonstrating their practical applicability.² Moreover, iron's diverse oxidation states (ranging from -2 to $+6$) and its ability to engage in multiple catalytic cycles, including oxidative addition, single-electron transfer, and Lewis acid activation, enable it to mediate a broad range of transformations. These features allow iron catalysis to facilitate the efficient construction of complex molecules, promoting the advancement of green chemistry and providing sustainable solutions for pharmaceutical synthesis and fine chemical production (Fig. 1a).^{3,4}

Difunctionalization of unsaturated bonds enables the simultaneous installation of two functional groups at adjacent positions, offering an efficient strategy for rapidly constructing complex molecular frameworks.^{5,6} This approach is particularly valuable in the synthesis of complex natural products and pharmaceuticals, as it significantly enhances molecular diversity (Fig. 1b). Early efforts primarily focused on the difunctionalization of activated alkenes; in contrast, unactivated alkenes often require additional directing groups to promote reactivity. The installation and subsequent removal of these directing groups introduce extra steps and reduce overall synthetic efficiency. With increasing mechanistic insights into alkene transformations, researchers have recognized that inherent

functional groups within substrates can serve as directing units, enabling efficient difunctionalization *via* distinct mechanistic pathways without the need for additional prefunctionalization. While nickel catalysis has traditionally dominated this field, the emergence of iron catalysis has opened new avenues for unsaturated bond difunctionalization.⁷ Although the first Fe-catalyzed coupling of aryl Grignard reagents was reported by Kharasch and Fields in 1941,⁸ it was not until 1971 that Kochi and Tamura demonstrated the first iron-catalyzed cross-coupling between Grignard reagents and vinyl halides.⁹ Only recently has Fe-catalyzed difunctionalization of unsaturated bonds gained significant attentions. Owing to its unique multivalent nature, iron can catalyze a variety of transformations involving alkenes, alkynes, and allenes with high efficiency and selectivity. By carefully tuning reaction conditions and ligand choice, iron catalysis enables the simultaneous introduction and transformation of functional groups, further advancing green and sustainable synthetic methodologies.

Given that comprehensive reviews on iron-catalyzed carbon-heteroatom bond formation are already available,¹⁰ this review does not cover the difunctionalization of unsaturated bonds involving both carbon-carbon and carbon-heteroatom bond formation. Instead, we focus specifically on iron-catalyzed



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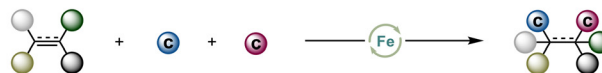


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a. Difunctionalization of unsaturated bonds via transition metal catalysis

$3d^1 4s^2$ Sc	$3d^2 4s^2$ Ti	$3d^3 4s^2$ V	$3d^4 4s^2$ Cr	$3d^5 4s^2$ Mn	$3d^6 4s^2$ Fe	$3d^7 4s^2$ Co	$3d^8 4s^2$ Ni	$3d^{10} 4s^1$ Cu	$3d^{10} 4s^2$ Zn
$4d^1 5s^2$ Y	$4d^2 5s^2$ Zr	$4d^4 5s^1$ Nb	$4d^5 5s^1$ Mo	$4d^5 5s^2$ Tc	$4d^7 5s$ Ru	$4d^8 5s^1$ Rh	$4d^{10}$ Pd	$4d^{10} 5s^1$ Ag	$4d^{10} 5s^2$ Cd
$5d^2 6s^2$ Hf	$5d^3 6s^2$ Ta	$5d^4 6s^2$ W	$5d^5 6s^2$ Re	$5d^6 6s^2$ Os	$5d^7 6s^2$ Ir	$5d^9 6s^1$ Pt	$5d^{10} 6s^1$ Au	$5d^{10} 6s^2$ Hg	



b. Synthesis of bioactive molecules via C=C bonds difunctionalization

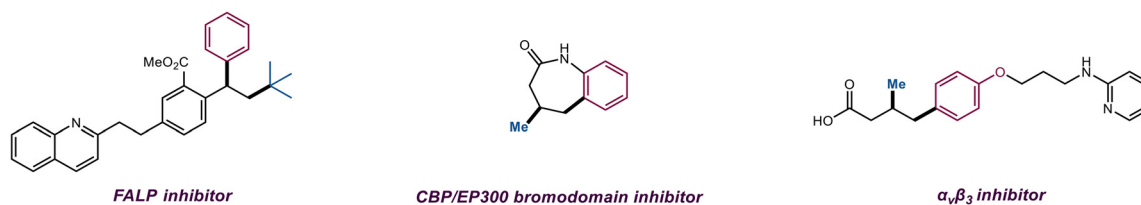
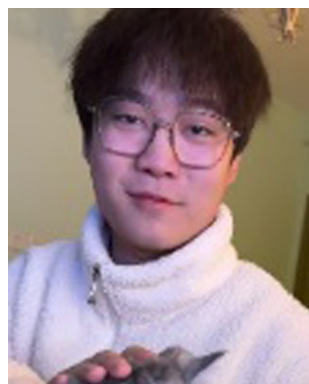


Fig. 1 Difunctionalization of unsaturated bonds via transition metal catalysis and its applications in the synthesis of bioactive molecules.

dicarbofunctionalization of alkenes, alkynes, and allenes. This review provides a comprehensive summary of recent advancements and mechanistic insights in this field, with particular emphasis on important applications in pharmaceutical synthesis and materials chemistry. Additionally, we also offer an in-depth discussion on reaction mechanisms and selectivity control. By summarizing the existing literature, we aim to provide both theoretical foundations and practical guidance for further developments in this area, ultimately contributing to the broader applications of green chemistry and sustainable synthetic strategies.

2. Iron-catalyzed dicarbofunctionalization of alkenes

Alkenes are abundant in bulk industrial feedstocks, and C=C bonds are commonly found in natural products, pharmaceuticals, and functional materials. Consequently, extensive efforts have been devoted to developing new synthetic methods for the functionalization of C=C bonds.¹¹ Among these, transition metal-catalyzed difunctionalization of olefins has emerged as a powerful strategy, enabling the transformation of readily



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available olefins into complex molecules with high step economy through diversification of the double bond. In particular, olefin dicarbofunctionalization has attracted significant attention, as it allows the simultaneous introduction of two carbon-based groups across the C=C bond, forming two new C-C bonds in a single step. Over the past few decades, substantial progress has been made in this area *via* iron catalysis, driven by intensive research efforts.

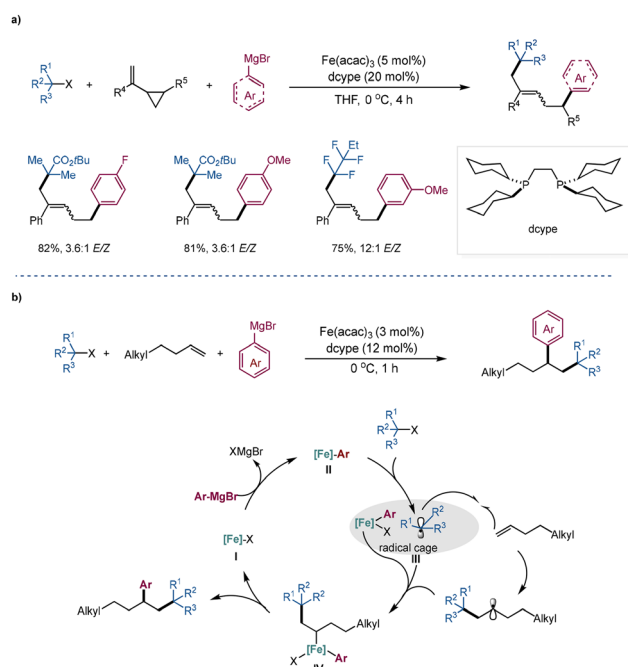
2.1. Dicarbofunctionalization of alkenes with organomagnesium reagents

In 2020, the Gutierrez group reported an iron-catalyzed radical ring-opening difunctionalization of vinylcyclopropanes (Scheme 1a).¹² By integrating computational and experimental studies, they matched the reactivity of alkyl radicals to undergo intra- and intermolecular radical cascade reactions *via* iron catalysis with both activated and unactivated vinylcyclopropanes, furnishing 1,5-alkylaryl products in moderate to good yields. Computational analysis revealed that tertiary alkyl radicals preferentially undergo intermolecular radical addition to vinylcyclopropanes (activation barrier: 14.0 kcal mol⁻¹) rather than radical coupling with iron species (activation barrier: 18.2 kcal mol⁻¹). The scope of this Fe-catalyzed three-component dicarbofunctionalization is broad with respect to the Grignard nucleophile. Nevertheless, certain limitations remain, including the high nucleophilicity of the Grignard reagent, which restricts compatibility with sensitive functional groups, and the lack of stereocontrol, leading to the formation of *E/Z* isomeric mixtures.

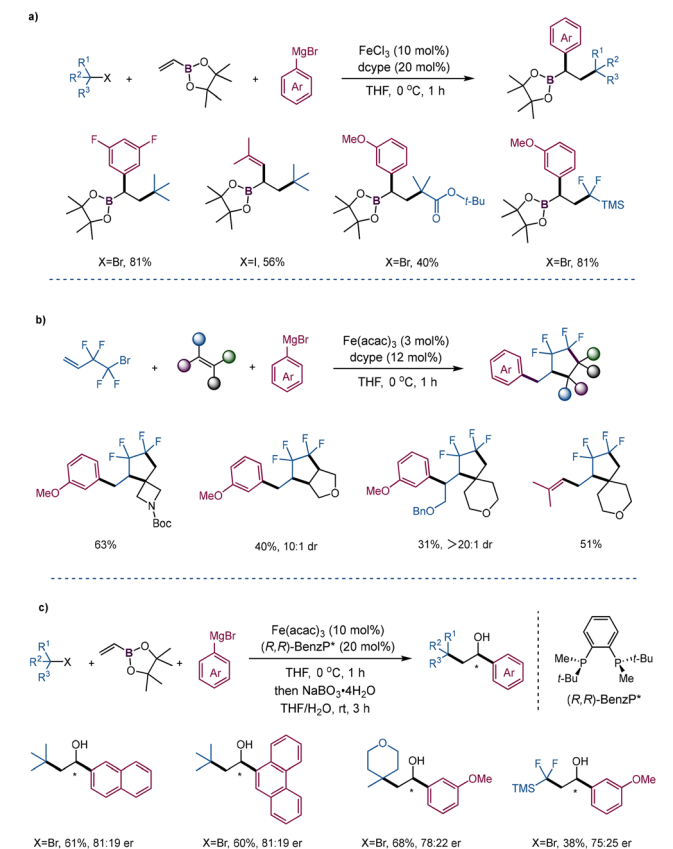
Selective transition metal-catalyzed three-component alkyl-arylation of unactivated alkenes remains a significant challenge,

particularly in the absence of electronically biased substrates or directing groups. In a notable advancement, the Gutierrez group reported the first iron-catalyzed three-component dicarbofunctionalization of unactivated terminal alkenes using both alkyl iodides and alkyl bromides in combination with aryl/alkenyl Grignard reagents (Scheme 1b).¹³ This protocol enables regio- and chemoselective 1,2-alkylarylation and 1,2-alkylvinyl-ation across a broad substrate scope. The success of this transformation relies on the use of Fe(acac)₃ as the catalyst and 1,2-bis(dicyclohexylphosphino)ethane (dcype) as the ligand, which together are crucial for promoting efficient three-component coupling. Despite this progress, the reaction requires a large excess of unactivated alkene (14 equivalents), likely due to the inherently low reactivity of such substrates. Additionally, aryl Grignard reagents must be introduced *via* slow addition, and the method displays limited compatibility with other primary and secondary alkyl halides, which tend to undergo undesired direct cross-coupling. As illustrated in Scheme 1b, the proposed mechanism involves initial activation of the alkyl halide by the iron catalyst to generate an alkyl radical and an aryl iron species (II). Owing to the steric hindrance and high energy barrier associated with direct cross-coupling between intermediate II and bulky alkyl radicals, the tertiary radical preferentially undergoes regioselective Giese-type addition to the alkene, forming a secondary alkyl radical intermediate. This less hindered radical is subsequently intercepted by the persistent aryl iron species, followed by reductive elimination from intermediate IV to afford the desired 1,2-dicarbofunctionalized product and regenerate the iron species I.

Organoboron compounds are widely utilized in medicine, materials science, and organic synthesis. Moreover, the palladium-catalyzed Suzuki-Miyaura cross-coupling reaction is among the most frequently employed methods in drug synthesis. Recently, alkenyl boronates have expanded the scope of multi-component reactions. In 2021, Gutierrez and Neidig reported an important iron-catalyzed three-component cross-coupling reaction involving alkyl halides, vinyl boronates, and aryl Grignard reagents, promoted by the bisphosphine ligand dcype (Scheme 2a).¹⁴ Concurrently, they developed a general method for iron-catalyzed multi-component cyclization cross-coupling that enables the rapid synthesis of fluorinated cyclic compounds and their derivatives (Scheme 2b). A wide range of fluoroalkyl radical precursors, such as those containing alkyl, ester, silyl, heteroaryl, phenoxy, and perfluoroalkyl proved compatible in this reaction, affording the desired 1,2-alkylfluorinated aryl organoboronates in good to excellent yields. The unique characteristics of the C-F bond, combined with the versatility of the C(sp³)-B bond and the practicality of this method, are anticipated to facilitate rapid access to valuable fluoroalkyl boronates for diverse synthetic applications. Mechanistic studies supported by DFT calculations indicate that the reaction initiates with a single-electron transfer between an Ar₂Fe species and the alkyl halide, generating a tertiary alkyl radical. This radical adds rapidly to the vinyl boronate to produce an α -boryl radical. The α -boryl radical then undergoes the cascade-cross-coupling reaction to yield the coupling product.



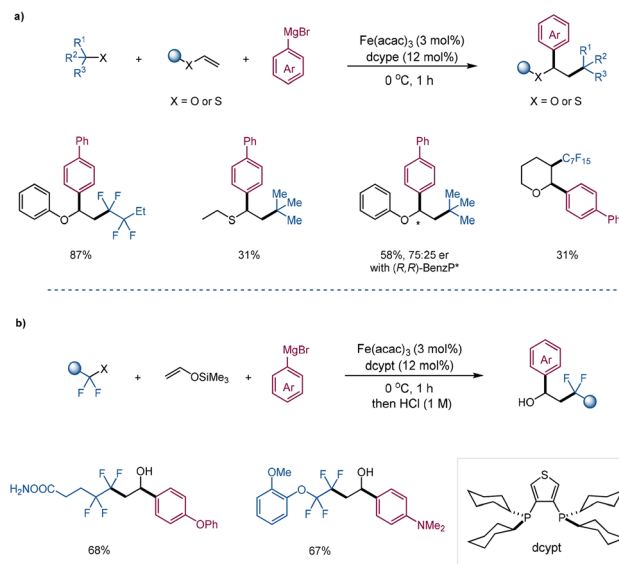
Scheme 1 Iron-catalyzed dicarbofunctionalization of vinylcyclopropanes and terminal alkenes with organomagnesium reagents.



Scheme 2 Iron-catalyzed dicarbofunctionalization of vinyl boronates with organomagnesium reagents.

In contrast to other transition-metal-catalyzed cross-coupling reactions, examples of Fe-catalyzed asymmetric cross-couplings involving sp^3 -hybridized coupling partners remain extremely rare. In 2023, the Gutierrez group made a significant advancement by achieving a general asymmetric multicomponent radical cross-coupling reaction (Scheme 2c).¹⁵ This novel process involves a range of difluoroalkyl halides, aryl Grignard reagents, and vinyl boronates. The authors identified $Fe(acac)_3$ and the chiral bisphosphine ligand (R,R) -BenzP* as a highly effective catalyst system, resulting in the desired multicomponent cross-coupling product with both good yield and excellent enantioselectivity. Notably, the reaction system displayed remarkable chemoselectivity, as the two-component cross-coupling and β -hydride elimination products were not observed. Instead, the primary byproducts were biaryl compounds, with a small amount of coupling product derived from the reaction between aryl Grignard reagents and vinyl boronates. This outcome underscores the unique efficiency and selectivity of the iron-catalyzed system, making it a promising approach for asymmetric multicomponent reactions.

The development of alkene difunctionalization reactions has predominantly focused on electron-deficient alkenes or alkenes with directing groups, which can effectively control reactivity and selectivity. However, catalytic methods for the selective 1,2-dicarbofunctionalization of electron-rich alkenes-offering straightforward access to modular synthesis of



Scheme 3 Iron-catalyzed dicarbofunctionalization of electron-rich alkenes with organomagnesium reagents.

complex scaffolds- remain scarce. In 2021, the Gutierrez group addressed this challenge by employing a bisphosphine ligand-iron catalytic system to achieve the 1,2-alkylarylation or 1,2-alkylalkenylation of alkenyl ethers and alkenyl sulfides (Scheme 3a).¹⁶ This reaction involves the use of tertiary alkyl halides and electron-donating and electron-withdrawing aryl/alkenyl Grignard reagents, demonstrating good functional group compatibility. Additionally, aryl alkenyl ethers and alkyl alkenyl ethers were smoothly transformed under the reaction conditions. Notably, the use of 4 equivalents of phenyl vinyl ether resulted in a reasonably good yield, contrasting with previous methods that required 14 equivalents of unactivated alkene substrates.¹³ This improvement is attributed to the Giese addition of alkyl radicals to vinyl ethers, which leads to more stable α -oxy radicals and drives the subsequent cross-coupling reaction. Furthermore, preliminary studies under optimized conditions suggest the potential for enantioselective coupling using (R,R) -BenzP* as the ligand.

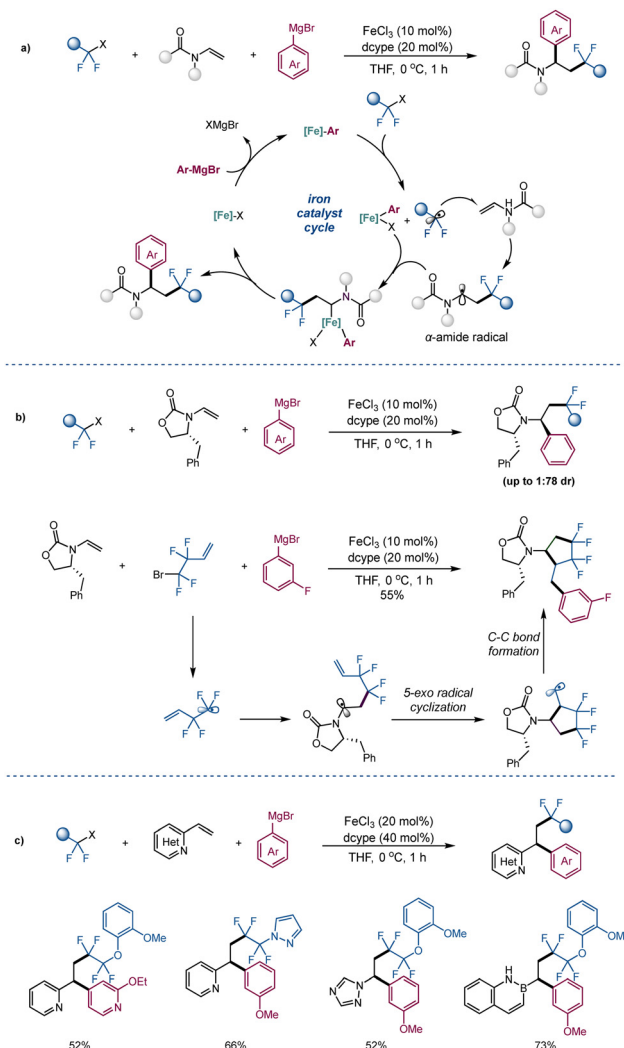
Enol silyl ethers, widely used as stable and easily prepared synthetic building blocks, have traditionally been limited to two-electron reaction systems. Despite their long-standing use as radical acceptors to generate transient α -silyloxy radicals *via* one-electron processes, the installation of a second C-C bond *via* a one-step selective radical/transition metal cross-coupling reaction has remained elusive. Building on their success with aryl (alkyl) enyl ethers, the Gutierrez group expanded this approach in 2023 by developing a fluoroalkyl arylation reaction system for enol silyl ethers (Scheme 3b).¹⁷ By utilizing only 2 equivalents of enol silyl ether, $Fe(acac)_3$, and the bisphosphine ligand (dcypt), they achieved the desired three-component coupling products with good yields. This system exhibited broad compatibility with a range of di-, tetra-, and perfluoroalkyl radical precursors, including those bearing alkyl, fluoroalkyl-rich chains, aryl, heteroaryl, aryl (thio)ethers, esters, and various

amides. This work significantly extends the scope of enol silyl ethers in radical/transition metal cross-coupling reactions, offering valuable strategies for the synthesis of fluorinated complex molecular architectures.

Enamides are valuable intermediates in organic synthesis due to their stability, reactivity, and presence as key structural components in many natural products and pharmaceuticals. Over the past few years, Zhang and Nevado have reported pioneering studies on nickel-catalyzed multi-component reactions involving enamides.¹⁸ However, these methods often suffer from drawbacks, such as long reaction times, high temperatures, and limited substrate scope. In contrast, the Gutierrez group discovered that bisphosphine-iron complexes could serve as highly effective catalysts for the fluoroalkylarylation of enamides *via* selective formation and cross-coupling of α -amide radicals.¹⁹ In these documents, they proposed that the *N*-containing group directly coordinated to the iron center, thereby leading to a (alkyl)Fe intermediate with high regioselectivity. In 2023, they reported an iron-catalyzed three-component reaction involving *N*-vinyl amides, fluoroalkyl halides, and aryl Grignard reagents. Despite its efficiency, this system was incompatible with heteroaryl Grignard reagents (Scheme 4a).

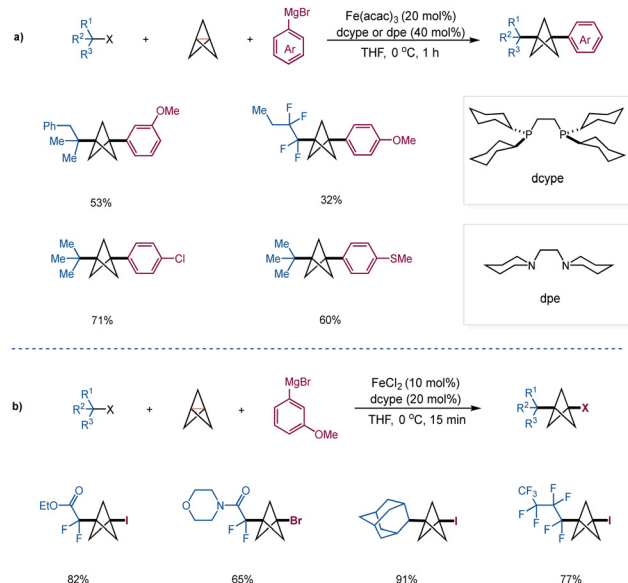
While the use of Evans' chiral oxazolidinones in asymmetric C-C bond formations is well-established in both academic and industrial settings, strategies utilizing chiral coupling partners to control diastereoselectivity in dicarbofunctionalization of alkenes *via* transition metal catalysis remain underdeveloped. In 2024, the Gutierrez group advanced this area by developing the first highly diastereoselective and broadly applicable three-component dicarbofunctionalization of alkenes bearing the Evans' chiral oxazolidinone using a bisphosphine-iron-catalyzed decoupled radical cross-coupling strategy (Scheme 4b).²⁰ This method employed readily available chiral oxazolidinones in reactions with alkyl halides and (hetero)aryl Grignard reagents, yielding products with good yields and exceptional diastereoselectivity (up to 1:78 dr). To confirm the formation of the α -amide radical intermediate, the authors conducted radical trapping experiments using a tetrafluoroalkyl halide with a terminal C=C bond. Under optimized conditions, this reagent led to the formation of a radical cyclic compound in 55% yield. These results validate the radical migration process, which forms the α -amide radical that is rapidly captured by the Ar-Fe species, facilitating diastereoselective C-C bond formation.

Building on this success, the Gutierrez group soon extended this concept to the fluoroalkyl(hetero)arylation of vinyl azaarenes (Scheme 4c),²¹ enabling the synthesis of unsymmetrical 1,1-bis(hetero)arylalkanes, including pyridine *N*-oxides, triazoles, pyrazines, carbazoles, indazoles, and 1,2-azaborines, with good efficiency. Interestingly, the efficiency of this reaction was found to be significantly influenced by the position of the nitrogen atom in the aromatic ring. The reactivity of these vinyl azaarenes suggests that when the nitrogen atom is in closer proximity to the reactive center, it promotes selective radical cross-coupling *via* weak interactions with the iron catalyst, thereby enhancing coupling efficiency.



Scheme 4 Iron-catalyzed dicarbofunctionalization of alkenes bearing nitrogen-directing groups with organomagnesium reagents.

In medicinal chemistry, bioisostere substitution strategies are widely employed to modulate the physicochemical properties of bioactive compounds. For example, replacing the benzene ring in bioactive molecules with C(sp³)-rich bicyclic hydrocarbons can yield novel structures with improved drug properties. The bicyclo[1.1.1]pentane (BCP) structure, an important building block in drug design, is commonly used to replace *para*-substituted arenes, *tert*-butyl groups, and alkynes due to its excellent bioisosteric properties.²² BCP substitution can enhance the pharmacokinetic properties of drug candidates by improving metabolic stability, water solubility, and membrane permeability. Despite the increasing number of synthetic methods for functionalizing BCPs through multicomponent reactions, catalytic cross-coupling reactions to form difunctionalized BCP-aryls remain rare, and the use of iron complexes as catalysts in this context has been largely unexplored. Furthermore, although C-F bonds are prevalent in bioactive compounds, one-pot, multicomponent cross-coupling methods for directly functionalizing [1.1.1]propellane to produce fluoroalkyl BCP-aryl scaffolds have not been



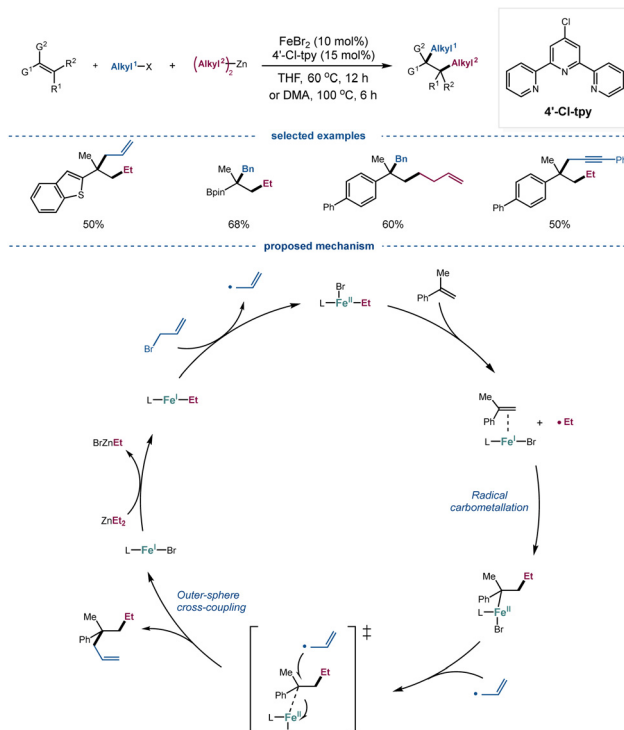
Scheme 5 Iron-catalyzed dicarbofunctionalization of BCPs with organo-magnesium reagents.

reported. In 2022, the Gutierrez group developed a conceptually distinct approach to access diverse (fluoro)alkyl BCP-aryls at low temperatures and in short reaction times.²³ This was achieved through an iron-catalyzed multicomponent radical cross-coupling reaction involving readily available (fluoro)alkyl halides, [1.1.1]propellane, and aryl Grignard reagents (Scheme 5a). Notably, the group also identified suitable reaction conditions for bisphosphine-Fe-catalyzed atom-transfer radical addition (ATRA) to [1.1.1]propellane, leading to the formation of the corresponding 1,3-fluoroalkyl-halo BCPs in excellent yields (up to 95%) (Scheme 5b). This breakthrough demonstrates the potential of iron-catalyzed processes for the efficient synthesis of functionalized BCP-aryl compounds, expanding the scope of bioisostere-based drug design.

2.2. Dicarbofunctionalization of alkenes with organozinc reagents

Most developments in the dicarbofunctionalization of alkenes focus on reactions involving the construction of both sp^2 - and sp^3 -hybridized carbons. In contrast to sp^2 -hybridized substrates, less-activated sp^3 -hybridized variants are more resistant to oxidative addition during cross-coupling reactions and are prone to side reactions, such as β -H elimination, which further complicate reaction systems.

In 2023, Koh's research group introduced an innovative approach using an $FeBr_2$ /tripyridine catalytic system to achieve the 1,2-dialkylation of olefins (Scheme 6).²⁴ This reaction employs alkyl zinc reagents and alkyl halides as alkylating agents to install two vicinal $C(sp^3)$ - $C(sp^3)$ bonds. Control experiments demonstrated that the reaction does not proceed *via* sequential alkylzincation, followed by trapping with alkyl halides, a typical mechanism found in other catalytic systems. Notably, only alkylzinc reagents provided good yields, while other organometallic reagents, including alkyl-MgBr, BuLi, and



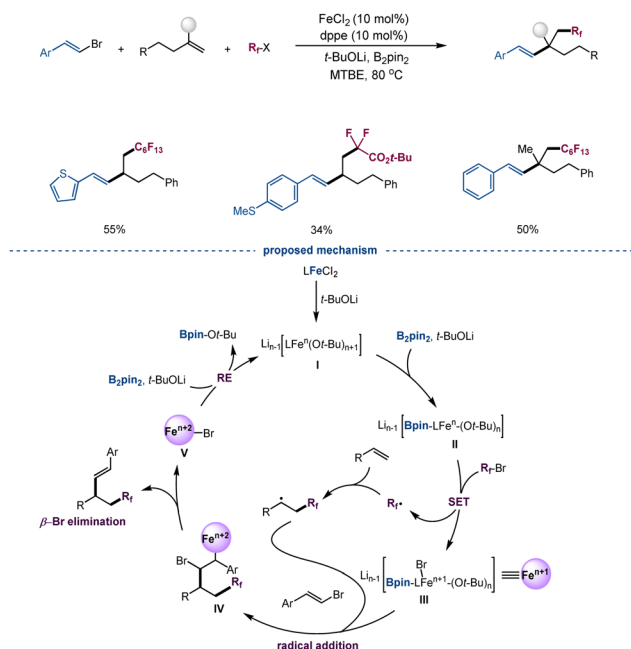
Scheme 6 Iron-catalyzed dicarbofunctionalization of alkenes with organozinc reagents.

$(alkyl)_3Al$, showed poor reactivity. To probe the generation of carbon-centered radical species in this system, radical clock experiments were conducted. Mechanistic studies revealed that the $Fe(II)Et$ complex initially form an iron-alkyl species *via* an alkyl radical carbometallation process. This species then undergoes alkylation with the alkyl halide through an outer-sphere coupling mechanism. Unlike existing iron-catalyzed systems that rely on stable alkyl radicals to drive the reaction, this catalytic system enables the use of unstable primary radicals, which undergo regioselective addition through the radical carbometallation pathway. This finding offers new insights for designing and developing iron-catalyzed reactions, expanding the scope of synthetic methods for functionalizing sp^3 -hybridized substrates.

2.3. Dicarbofunctionalization of alkenes *via* reductive coupling

Although iron-catalyzed Kumada/Negishi-type dicarbofunctionalization of alkenes has been well-documented in recent years,^{12–24} the use of iron catalysis for reductive coupling in multicomponent reactions remains underexplored. One of the main challenges hindering the development of these reactions is the difficulty in identifying a catalytic system capable of generating highly active iron species for electrophile activation, a crucial step in facilitating these transformations.

In 2024, the Feng group reported the first diboron-promoted fluoroalkylalkenylation of unactivated alkenes under iron catalysis, using $dppe$ as the ligand and B_2pin_2 as the reductant (Scheme 7).²⁵ Density functional theory (DFT) calculations



Scheme 7 Iron-catalyzed regioselective reductive fluoroalkylalkenylation of unactivated alkenes.

revealed that non-fluoroalkyl radicals exhibit comparable energy barriers and reaction rates when adding to both alkenes and alkenyl bromides, potentially leading to poor chemoselectivity. In contrast, fluoroalkyl radicals add to alkenes at a significantly faster rate than to alkenyl bromides. This distinction highlights the importance of selecting an appropriate radical precursor to achieve high chemoselectivity in alkyl-alkenylation and suppress competing side reactions. This reaction involved the coupling of aryl alkenyl bromides, unactivated alkenes, and fluoroalkyl halides under mild conditions, proceeding through a free radical cascade addition pathway. The process afforded stereodefined homoallylic fluoroalkanes with high efficiency and regioselectivity. Notably, the reaction displayed good functional group compatibility, and the desired products were obtained in good to excellent yields without the need for directing groups. However, nonfluoroalkyl halides were unsuitable for this three-component reductive coupling, as they predominantly underwent a two-component coupling with the aryl alkenyl bromides. The reaction was also successful with disubstituted terminal alkenes, yielding the three-component product in 50% yield. Unfortunately, the catalytic system showed poor reactivity with internal alkenes.

Based on radical inhibition studies and other control experiments, a plausible mechanism for the iron-catalyzed fluoroalkylalkenylation was proposed, as shown in Scheme 7. The reaction begins with the treatment of $\text{Fe}(\text{OAc})_2$ with $t\text{-BuOLi}$ to form the iron-ate complex **I**. This complex reacts with B_2pin_2 in the presence of base to generate the iron-boryl species **II**. The resulting complex **II**, possessing strong reducing ability, initiates a single-electron transfer (SET) process with the fluoroalkyl bromide, generating a fluoroalkyl radical and the iron species **III**. The fluoroalkyl radical then attacks the internal alkene, forming an alkyl radical intermediate. This intermediate is subsequently captured by the alkenyl bromide and iron species **III**, yielding intermediate **IV**. Finally, the desired three-component reductive coupling product is obtained through $\beta\text{-Br}$ elimination.

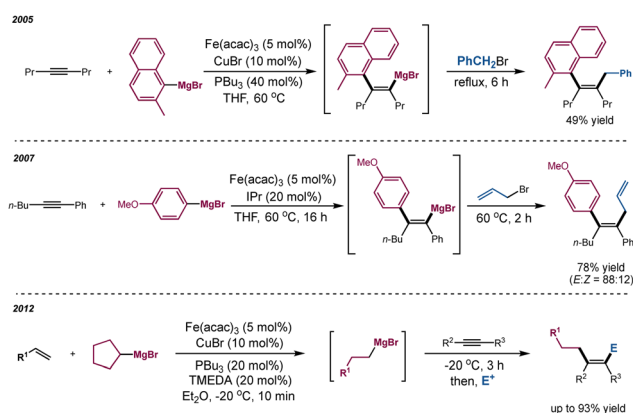
forming an alkyl radical intermediate. This intermediate is subsequently captured by the alkenyl bromide and iron species **III**, yielding intermediate **IV**. Finally, the desired three-component reductive coupling product is obtained through $\beta\text{-Br}$ elimination.

3. Iron-catalyzed dicarbofunctionalization of alkynes

Alkynes with a carbon-carbon triple bond are important in organic chemistry as versatile building blocks for synthesizing various compounds, including pharmaceuticals, materials, and fuels, due to their unique reactivity and ability to be transformed into other functional groups. Therefore, great efforts have been made to develop new synthetic methods to utilize such compounds. Among the developed methods, transition metal-catalyzed difunctionalization of alkynes enables simple preparation of complex molecules by diversifying carbon-carbon triple bonds with high step economy.²⁶ It can build complex carbon skeletons by assembling two carbon groups on the carbon-carbon triple bond and forming two new C-C bonds in a one-step reaction. However, this field has received little attention until recently when a few research groups have made remarkable progress in this field.

3.1. Dicarbofunctionalization of alkynes with organomagnesium reagents

Carbometallation of alkynes is a highly effective strategy for synthesizing multisubstituted alkenes, as the resulting alkenyl-metal intermediates can be further transformed with electrophiles in a stereoselective manner. In 2005, the Hayashi group reported the arylmagnesiation of non-functionalized alkynes using a bimetallic iron/copper catalytic system (Scheme 8).²⁷ In this transformation, both metal catalysts are crucial: omission of $\text{Fe}(\text{acac})_3$ resulted in complete recovery of the alkyne, while the absence of CuBr led to a diminished yield (26%) of the coupling product. The *E*-geometry product indicates that arylmagnesiation took place with *syn*-selectivity. Mechanistic studies indicated that the copper catalyst primarily promotes metal exchange between the alkenyliron intermediate and the



Scheme 8 Iron-catalyzed dicarbofunctionalization of internal alkynes with organomagnesium reagents.

aryl Grignard reagent. The synthetic utility of the arylmagnesian products was demonstrated in a subsequent one-pot reaction with benzyl bromide to afford a tetrasubstituted alkene in 49% yield.

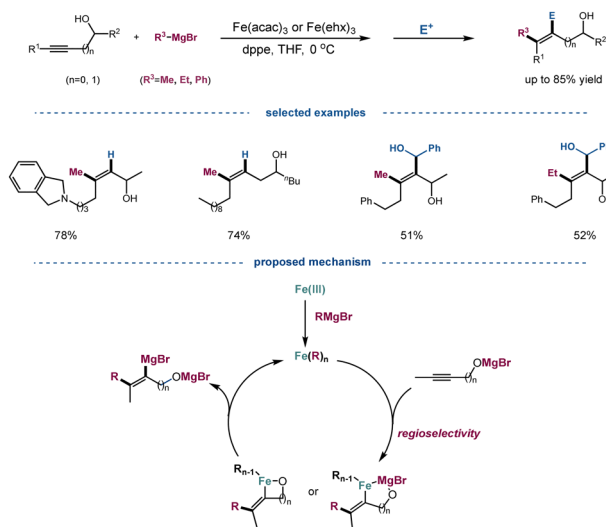
Soon after, the same research group improved the arylmagnesian products by introducing a catalytic amount of an N-heterocyclic carbene ligand, which enabled the formation of trisubstituted alkenylmagnesiums in high yields without the need for copper as a cocatalyst (Scheme 8).²⁸ These alkenylmagnesium intermediates were then efficiently converted into tetrasubstituted alkenes upon treatment with allyl bromide. In 2012, leveraging their earlier findings, the Hayashi group further extended their methodology by employing an iron-copper bimetallic system in a one-pot process (Scheme 8).²⁹ In this protocol, terminal alkenes underwent alkene-alkyl Grignard exchange with alkyl magnesium bromide to generate alkyl Grignard reagents, which then participated in alkenylmagnesian products to furnish the desired products in good yields. This progression of studies not only highlights the evolving utility of iron- and copper-catalyzed carbometalation of unsaturated bonds but also underscores the importance of catalytic system design in achieving efficient and stereoselective multi-component synthetic transformations.

Carbometalation of terminal alkynes is a broadly utilized method for constructing complex molecules. However, early investigations revealed that the carbometalation of electronically unbiased internal alkynes is typically slow and suffers from poor regioselectivity. In 2006, the Ready group addressed these limitations by introducing polar functional groups near the alkyne, thereby modulating both the rate and regioselectivity of the carbometalation reaction (Scheme 9).³⁰ They successfully demonstrated an iron-catalyzed alkenylmagnesian products of propargylic and homopropargylic alcohols. Under these conditions, propargyl and homopropargyl alcohols reacted with alkyl or aryl Grignard reagents to afford allyl and homoallyl alcohols

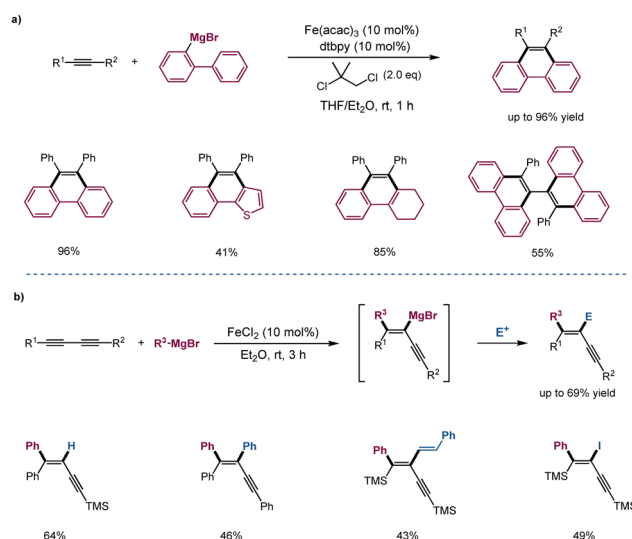
with excellent stereoselectivity and good regioselectivity. A key advantage of this approach is that the resulting vinyl Grignard intermediate can be intercepted with a range of electrophiles, enabling the efficient synthesis of tetrasubstituted allylic alcohols. Mechanistically, it was proposed that the hydroxyl group directly coordinated to the iron center, thereby promoting an alkoxide-directed carbometalation that led to a (vinyl)Fe intermediate with high regioselectivity. Alternatively, the reaction might be facilitated by the association of iron with magnesium. The ensuing (vinyl)Fe(R) species then underwent transmetalation with the Grignard reagent, furnishing the final carbometalated product. This work highlights the significant role that substrate modification and catalyst design can play in enhancing both the efficiency and selectivity of carbometalation processes.

In 2011, the Nakamura group reported an iron-catalyzed [4+2] cyclization of internal or terminal alkynes with 2-biaryl, 2-heteroarylphenyl, or 2-alkenylphenyl Grignard reagents using 1,2-dichloro-2-methylpropane as the oxidant (Scheme 10a).³¹ This transformation afforded 9-substituted or 9,10-disubstituted phenanthrenes and related analogues in moderate to good yields. Remarkably, when 1,3-diynes were used as substrates, cyclization occurred at both acetylene units to provide biphenylene derivatives. The proposed mechanism involves either the carbometalation of the internal alkyne followed by C–H bond activation or the formation of a biphenyl ferracycle intermediate. Building on this work, in 2014 the same research group achieved a chemo-, regio-, and stereoselective carbomagnesian products of 1,3-diynes with aryl Grignard reagents under iron catalysis (Scheme 10b),³² which enabled the efficient synthesis of polysubstituted 1,3-enyne derivatives. These studies collectively underscore the versatility of iron catalysis in constructing complex polycyclic frameworks *via* efficient dicarbofunctionalization of alkynes.

Monocarbometalation of alkynes is one of the most developed methods for the stereoselective and regioselective synthesis of vinyl lithium, magnesium, aluminum, and zinc reagents. Ideally, the



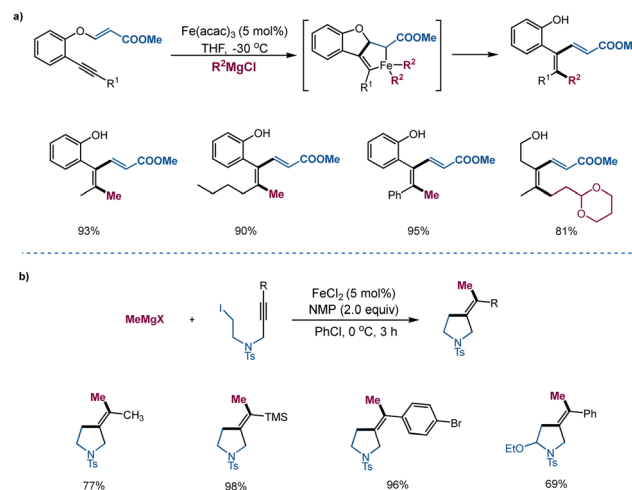
Scheme 9 Iron-catalyzed dicarbofunctionalization of propargylic and homopropargylic alcohols with organomagnesium reagents.



Scheme 10 Iron-catalyzed dicarbofunctionalization of internal alkynes.

vinyl metal intermediate produced in such a reaction could undergo additional carbometallation with alkynes to yield di-, tri-, or even polyenyl metal reagents. However, achieving precise control over the degree of alkyne insertion remains a significant challenge in multicarbometallation processes. In 2016, the Deng group addressed this challenge by developing an iron(II)-N-heterocyclic carbene (NHC)-catalyzed double-carbometallation of internal asymmetric alkynes using alkyl Grignard reagents (Scheme 11).³³ The reactions employing the more steric demanding NHC ligands, such as IMes and IPr, or using $\text{Fe}(\text{acac})_3$, $\text{FeCl}_2/\text{PPh}_3$, FeCl_2 , and $\text{Fe}(\text{acac})_3$ as catalysts mainly afford polyalkynes. This method affords highly alkyl-substituted 1,3-dienyl magnesium reagents with excellent regio- and stereoselectivity, which can be readily quenched with D_2O , I_2 , or other electrophiles to afford the corresponding functionalized products. Mechanistic investigations suggest that low-coordinate alkyl iron(II)-NHC species participate in the reaction as the key intermediate.

In 2016, Fürstner reported a groundbreaking iron-catalyzed cascade reaction that merges cycloisomerization and cross-coupling, enabling the stereocontrolled construction of two vicinal C–C bonds *via* metallacyclic intermediates (Scheme 12a).³⁴ This strategy effectively overcomes the challenges associated with the stereospecific synthesis of tetrasubstituted olefins in functionalized 1,3-dienes, through either inert C–X (X = O, N) or activated C–C bond cleavage. Operating under a ligand-free system with $\text{Fe}(\text{acac})_3$ and alkyl Grignard reagents, the reaction circumvents the β -hydrogen elimination pathway by exploiting the redox flexibility of the iron center and the formation of ate complexes. Subsequent investigations confirmed that the iron-catalyzed system was broadly applicable to both aliphatic and electron-deficient substrates while maintaining high stereoselectivity. The reaction was proposed to proceed *via* metallacyclic intermediates, and the iron center has gained ate character before reductive elimination occurs.

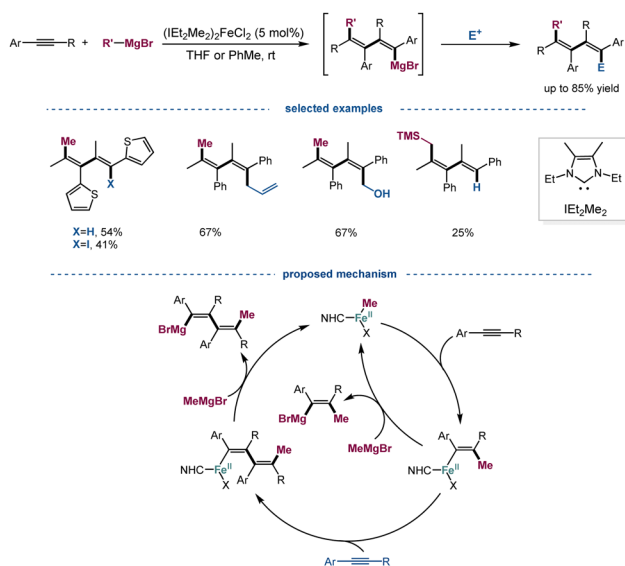


Scheme 12 Iron-catalyzed cyclization and dicarbonylation of alkynes.

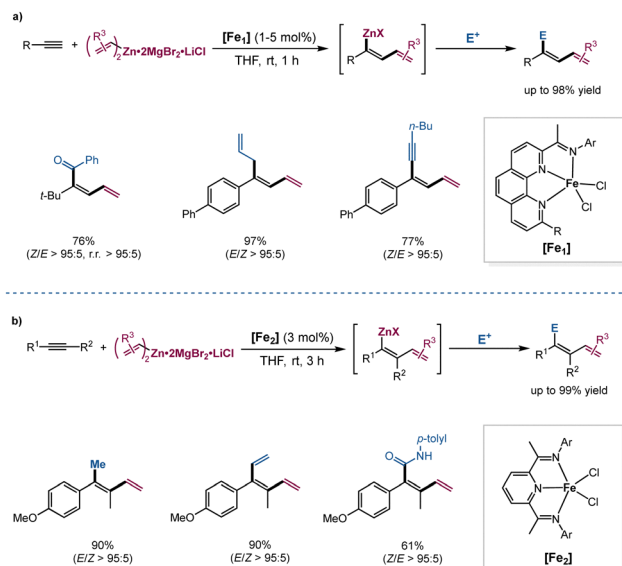
Building on these advances, in 2019 the Kang group also developed an iron-catalyzed, chemoselective cyclization and cyclomethylation strategy employing alkyl Grignard reagents (Scheme 12b).³⁵ Using FeCl_2 as the catalyst, they successfully achieved 5-exo cyclizations to synthesize alkenyl-containing pyrrolidines, tetrahydrofurans, and carbocyclic compounds. This study addressed critical challenges in iron catalysis, including the suppression of β -hydrogen elimination and the selective control of cyclization *versus* alkyl transfer. Mechanistic studies suggest that the alkyl-iron species activate alkyl iodides through a single-electron transfer (SET) pathway to generate free alkyl radical intermediates, which then participate in the subsequent cyclization process.

3.2. Dicarbonylation of alkynes with organozinc reagents

Although carbocationization of terminal alkynes offers a promising approach for generating alkenylzinc reagents, versatile intermediates widely used in organic synthesis, challenges remain in achieving high chemo-, regio-, and stereoselectivity. In 2021, the Zhu group addressed these issues by developing a method for vinylzincation using a 1,10-phenanthroline-imine-ligated iron catalyst (Scheme 13a).³⁶ In contrast, when FeCl_2 was used as the catalyst in the absence of the ligand, substrate conversion was incomplete, and the reaction exhibited poor selectivity, producing a mixture of carbocationization products along with polymerized alkyne derivatives and unidentified byproducts. This protocol efficiently converts terminal alkynes into a diverse array of conjugated alkenylzinc compounds in high yields, with excellent selectivity and high *cis*-stereoselectivity. The reaction exhibits broad functional group tolerance and accommodates a wide substrate scope, including aryl, alkenyl, and alkyl acetylenes bearing various functionalities. Mechanistic investigations, including reduction experiments employing (*p*-Tol)MgBr and di(1-phenylvinyl)zinc, suggest that an Fe(I) species may be formed under these catalytic conditions. Further studies indicate that the crowded reaction pocket of



Scheme 11 Iron(II)-catalyzed double-carbometallation of internal asymmetric alkynes using alkyl Grignard reagents.

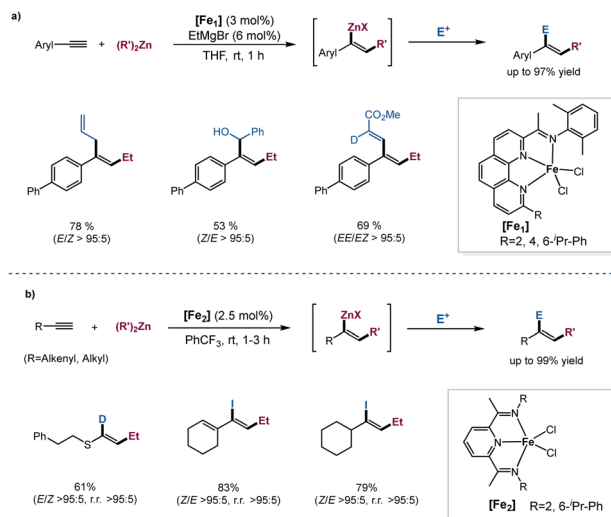


Scheme 13 Iron-catalyzed dicarbofunctionalization of alkynes with alkenylzinc reagents.

the iron-1,10-phenanthroline-imine catalyst promotes efficient alkenyl group transfer to the alkyne while suppressing undesired substitution reactions at the terminal C–H bond. These findings highlight the potential of this catalytic system for selective vinylzincation and its utility in constructing complex, conjugated molecules.

Regioselectivity in the vinylzincation of unsymmetrical dialkyl acetylenes represents a formidable challenge due to the similar electronic and steric properties of the two alkyl substituents. In 2021, the Zhu group advanced this field by demonstrating an iron-catalyzed alkenylzincation of internal alkynes using a pyridinediimine-ligated iron catalyst (Scheme 13b).³⁷ Notably, this catalytic system achieves highly selective vinylzincation for both unfunctionalized internal alkynes and unsymmetrical diaryl/dialkyl acetylenes, which have proven challenging with other catalytic methods. The reaction exhibits impressive reactivity (with turnover numbers up to 11 500), high selectivity, broad functional group tolerance, and a wide substrate scope. Importantly, the resulting alkenylzinc species serve as versatile intermediates that can be transformed into a series of multi-substituted conjugated dienes through a variety of subsequent reactions.

The selective synthesis of multi-substituted alkenes, particularly with precise control over stereochemistry, remains a formidable challenge. Building on these advances made in his group, in 2022, Zhu introduced an efficient method for the alkylzincation of terminal alkynes using either 1,10-phenanthroline-imine-ligated or pyridinediimine-ligated iron catalysts (Scheme 14).³⁸ This reaction furnishes a diverse array of alkenylzinc compounds in high yield, featuring excellent anti-Markovnikov selectivity and high *cis*-stereoselectivity. Moreover, by simply varying the iron complex, the alkylzincation of a range of acetylenes containing aryl-, alkenyl-, alkyl-, or heteroatom-substituted can be readily achieved. Importantly,



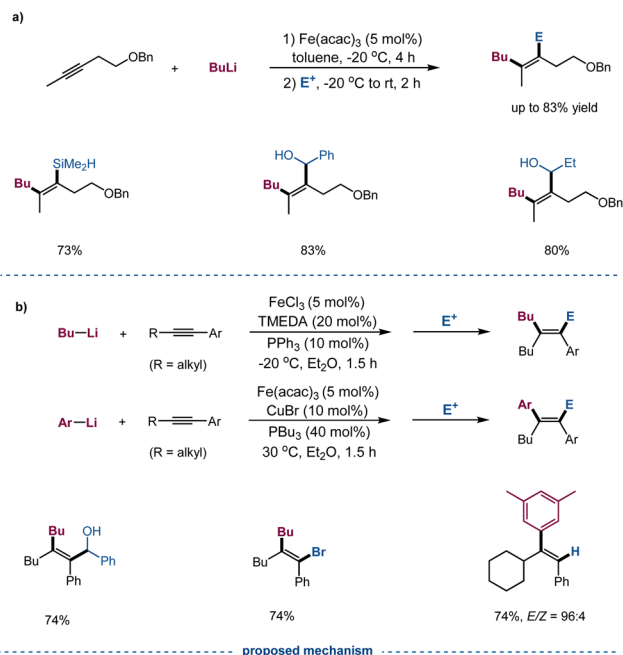
Scheme 14 Iron-catalyzed dicarbofunctionalization of terminal alkynes with alkylzinc reagents.

the resulting alkenylzinc products serve as versatile intermediates that can be further functionalized in one-pot Negishi coupling reactions, enabling the efficient synthesis of trisubstituted olefins. This transformation represents a competitive alternative to traditional methods such as the Wittig reaction, Julia–Lythgoe olefination, and olefin metathesis for the selective synthesis of trisubstituted alkenes.

3.3. Dicarbofunctionalization of alkynes with organolithium reagents

Organolithium reagents have garnered significant attention due to their ready availability and dual functionality as powerful nucleophiles for carbon–carbon bond formation and efficient metal transfer agents in transmetalation processes. However, their high basicity poses considerable challenges in alkylolithiation of alkynes. In particular, organolithium reagents readily deprotonate terminal alkynes, and even when addition is facilitated by intramolecular processes or heteroatom-directed groups (*e.g.*, alkoxy or amino substituents), side reactions at the propargyl position are common. This deprotonation issue is particularly pronounced for alkynes bearing simple alkyl substituents. Moreover, the synthetic utility of alkynes in carbolithiation reactions is further limited by the narrow substrate scope and the propensity of the resulting vinylolithium intermediates to isomerize.

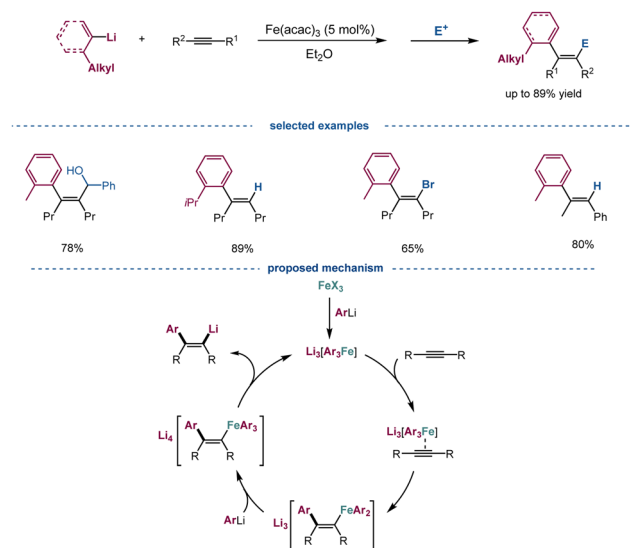
To overcome these challenges, the Hosomi group made a breakthrough in 2001 by developing an iron-catalyzed carbolithiation of alkynes using inexpensive iron salts (Scheme 15a).³⁹ This methodology enabled both regioselective and stereoselective carbolithiation. Notably, increasing the loading of butyllithium from 8 to 12 equivalents improved the yield of the desired product from 63% to 89%, suggesting that an iron–ate complex may be the active catalytic species. Although this system effectively addressed the alkylation compatibility of methyl-substituted alkynes, its scope was limited by the requirement for heteroatom directing groups (such as ether or amine).



Scheme 15 Iron-catalyzed dicarbofunctionalization of alkynes with alkyl-lithium reagents.

This limitation was subsequently overcome by the Hayashi group in 2009 (Scheme 15b).⁴⁰ They demonstrated that the reaction could proceed efficiently in the presence of TMEDA and PPh₃ as ligands, affording the corresponding alkylated products in good yields and with high stereoselectivity. Moreover, their studies revealed that the initial product is formed as a *syn*-adduct, which isomerizes to the more thermodynamically favored *trans*-adduct under the reaction conditions. This new iron catalytic system successfully achieves the efficient alkylation of alkynes without the need for heteroatom directing groups, thereby significantly broadening the scope and applicability of carbon metalation strategies in organic synthesis.

Although numerous methods exist for the carbometallation of alkynes using arylmetallic reagents, these strategies often require harsh conditions and are subject to stringent substrate restrictions. This is particularly problematic for dialkyl acetylenes lacking heteroatom directing groups, where the metallation process presents significant challenges. In 2011, the Hayashi group demonstrated that Fe(acac)₃ could catalyze the highly selective addition of aryllithium and vinylolithium reagents to alkynes under mild conditions without the need for additional additives (Scheme 16).⁴¹ Notably, aryllithium and vinylolithium reagents bearing an alkyl substituent at the



Scheme 16 Iron-catalyzed dicarbofunctionalization of alkynes with aryl/alkenyl lithium reagents.

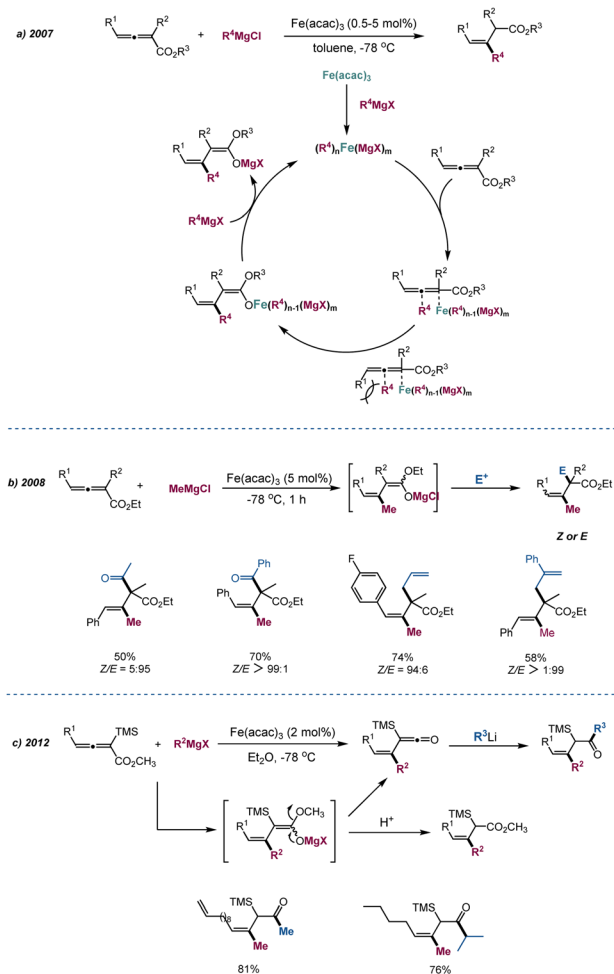
ortho- or *cis*-position, respectively, reacted smoothly, whereas substrates without these bulky substituents showed little to no reactivity. This observation was rationalized by proposing that substrates lacking a bulky group form an iron-ate complex, Li₄[Ar₄Fe], which was less prone to coordinate with the alkyne. In contrast, the presence of a bulky substituent favors the formation of a less congested Li₃[Ar₃Fe] complex, which readily coordinates with the alkyne and facilitates the coupling reaction.

4. Iron-catalyzed dicarbofunctionalization of allenes

Allenes are important in organic chemistry because they are versatile building blocks for the synthesis of complex molecules, found in natural products and pharmaceuticals, and serve as unique synthons in modern organic synthesis and bioactive molecule design. Therefore, great efforts have been made to develop new synthetic methods to utilize such compounds. Among the methods that have been developed, transition metal-catalyzed difunctionalization of allenes enables simple preparation of complex molecules by diversifying C=C double bonds with high atom and step economy.⁴² It can build complex carbon skeletons by forming two C-C bonds in a one-step reaction, while retaining double bond functional groups in the product, which can be used for later derivatization. However, this field has received little attention until recently, when a few research groups have made remarkable progress in this field.

4.1. Dicarbofunctionalization of allenes with organomagnesium reagents

The migration of the C=C double bond to the conjugated α,β -position complicates the synthesis of β,γ -unsaturated



Scheme 17 Iron-catalyzed dicarbofunctionalization of allenes with organomagnesium reagents.

alkenoates. In 2007, the Ma group reported a highly regio- and stereoselective conjugate addition of Grignard reagents to 2,3-alkenoates under iron catalysis (Scheme 17a and b).^{43,44} Using $Fe(acac)_3$ as the catalyst, primary/secondary alkyl, phenyl, or vinyl Grignard reagents selectively added to the β -position of allenoates in toluene at -78°C , yielding β,γ -unsaturated alkenoates. Mechanistic studies revealed that $Fe(acac)_3$ formed an iron-magnesium ate complex with the Grignard reagent, facilitating the conjugate addition to the α,β -double bond of the allenoate, resulting in 1,3-dienolate magnesium intermediates. The stereoselectivity arises from steric repulsion between the R^1 and R^4 groups in the transition state. Importantly, the resulting magnesium dienolate intermediates can undergo downstream transformations, such as reaction with aldehydes, methyl chloroformate, or acetyl chloride, to form a second C–C bond with high efficiency.

While these advances in dienolate transformations were significant, there had been no report on a stereocontrolled reaction of the same metal dienolate with different electrophiles to afford β,γ -unsaturated (*E*)- or (*Z*)-alkenoates with high regio- and stereoselectivity. In 2008, the Ma group addressed

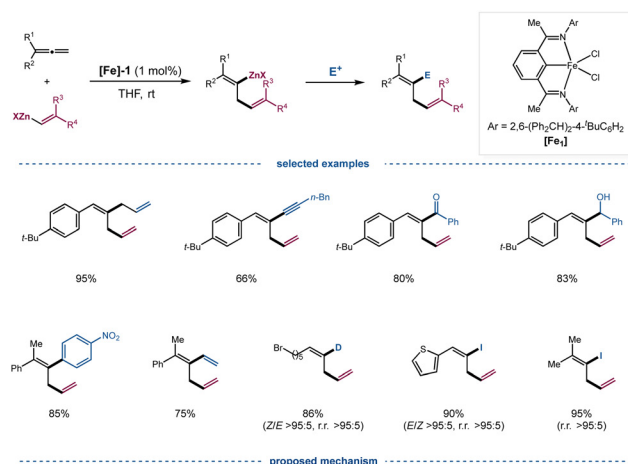
this gap by reporting a controllable, highly stereoselective reaction of iron-catalyzed coupling between 2,3-allenoates and Grignard reagents with various electrophiles. The stereochemistry of the product could be precisely controlled by adjusting the reaction conditions and catalysts. They observed efficient and controllable stereoselective α -acylation or α -allylation reactions of the *in situ* generated magnesium dienolate intermediate, formed from the Fe-catalyzed conjugate addition of 2,3-allenoates with Grignard reagents. These magnesium dienolate intermediates then reacted with acyl chloride, allyl carboxylic acid esters, allylic acetate (catalyzed by $Pd(PPh_3)_4$), or allylic bromide (catalyzed by CuI) to afford alkyl-3(*Z* or *E*)-enoates with high stereoselectivity. In 2009, the Ma group extended their research to achieve a highly regio- and stereoselective conjugate addition reaction catalyzed by $FeCl_3 \cdot 6H_2O$, allowing the synthesis of multi-substituted β,γ -unsaturated alkenoates.⁴⁵

In 2012, the same group reported a highly efficient reaction of 2-trimethylsilyl-2,3-allenoate with Grignard reagents, catalyzed by iron, to synthesize trimethylsilyl vinyl ketenes with high regio- and stereoselectivity (Scheme 17c).⁴⁶ This reaction was performed in Et_2O at -78°C under $Fe(acac)_3$ catalysis. Mechanistic studies indicated that Grignard reagents and allenoates underwent conjugate addition to form a 1,3-dienol magnesium intermediate, which then eliminated methoxymagnesium salt to yield enones. The trimethylsilyl group plays a crucial role in the transformation. Notably, these products have been applied in the synthesis of stereodefined unsaturated enones, trienes, and polysubstituted phenols.

4.2. Dicarbofunctionalization of allenes with organozinc reagents

The addition of nucleophiles to allenes typically occurs at the electron-deficient central carbon atom due to the structural characteristics of allenes. In 2025, the Zhu group discovered an intriguing variation of this process, where the nucleophilic addition of alkenylzinc reagents to allenes catalyzed by the iron complex, resulting in the electrophilic metal zinc being positioned on the central carbon atom of the allene (Scheme 18).⁴⁷ Moreover, control experiments showed that the $FeCl_2$ was ineffective in the vinylzincation reaction, and no desired products were found. This method exhibits unconventional regioselectivity, with the reaction proceeding under mild conditions and generally providing high yields (typically $> 80\%$), excellent regioselectivity, high *syn*-stereoselectivity, and excellent functional group tolerance. This method enables the synthesis of a range of 1,4-diene zinc reagents, which are highly versatile and readily undergo further transformations, ultimately yielding functionalized 1,4-dienes that are challenging to obtain *via* other synthetic routes.

To elucidate the mechanistic details of this iron-catalyzed allene alkenylzincation reaction, several control experiments were conducted. The configuration of the organozinc product was confirmed by NMR spectroscopy, and radical inhibition reaction also ruled out a free radical mechanism. Stoichiometric reduction experiments and the evaluation of $Fe(0)$ species activity

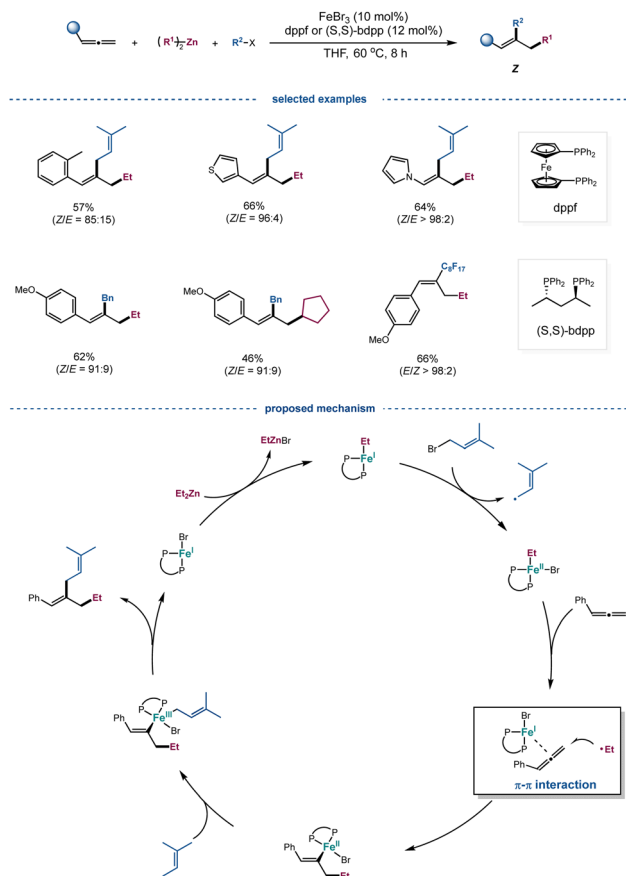


Scheme 18 Iron-catalyzed dicarbofunctionalization of allenes with organozinc reagents.

suggest that the reaction was likely promoted by Fe(0) catalysis. Mechanistic analysis revealed that the electron-donating ability of Fe(0) played a crucial role in this catalytic system. Upon coordination with the allene, iron transfers an electron to the allene through π back-bonding, thereby activating the coordinated double bond. This coordination effect is more pronounced on the central carbon atom than on the terminal carbon atom of the allene, potentially inverting the electrophilicities of the two carbons. The increased electrophilicity of the terminal carbon atom makes it more susceptible to nucleophilic attack, explaining the observed abnormal regioselectivity. These findings suggest that a unique iron electron-delocalization mechanism underlies the unusual regioselectivity in this reaction.

In organic synthesis, the stereoselective construction of trisubstituted alkenes remains a central challenge in the synthesis of asymmetric olefins. This is particularly true for the controlled formation of *Z*-isomers, which are often thermodynamically disfavored due to significant steric repulsion between substituents and the typically small energy difference between *Z/E* isomers. This intrinsic contradiction is especially pronounced in the synthesis of acyclic trisubstituted alkenes. Currently, efficient and general methods for the synthesis of stereodefined *Z*-alkenes remain scarce.

In 2025, the Koh group reported an iron-catalyzed multi-component strategy that enables the dialkylation of allenes



Scheme 19 Iron-catalyzed dialkylation of allenes with organozinc reagents.

using dialkylzinc reagents and alkyl halides (Scheme 19).⁴⁸ This method provides a series of trisubstituted alkenes with excellent regioselectivity and *Z*-selectivity. The synthetic utility of this protocol was further demonstrated by the efficient synthesis of a glucosylceramide synthase inhibitor. Mechanistic studies supported by computational analysis revealed a kinetically controlled pathway that contrasts with conventional metal-catalyzed allene functionalization mechanisms. Specifically, the reaction proceeded through a radical-mediated alkylation preferentially at the sterically less hindered face of the allene, followed by a reductive elimination step to achieve inner-sphere alkylation. Notably, π - π interactions between the aryl group on the allene and the ligand on the catalyst lower the activation energy, thereby enhancing the stereoselectivity of the transformation. This strategy provides valuable insights into the design of kinetically controlled transformations of allenes and other π -systems.

5. Summary and outlook

In recent years, iron-catalyzed dicarbofunctionalization of unsaturated carbon-carbon bonds has emerged as a powerful strategy for constructing complex frameworks. This review systematically summarizes the distinct mechanistic pathways associated with various organometallic reagents including

RMgX, RZnX, and RLi in iron-catalyzed systems for the dicarbofunctionalization of alkenes, alkynes, and allenes, with a focus on their applications and underlying reaction mechanisms.

For alkene transformations, three-component coupling systems have established relatively mature radical-mediated pathways. In particular, arylmagnesium reagents preferentially form aryliron species that capture alkyl radical intermediates, whereas the coupling efficiency of organozinc reagents is limited by reactivity mismatches with electrophilic partners. Despite progress, the application of free-radical cascade strategies remains constrained by the limited scope of compatible electrophiles.

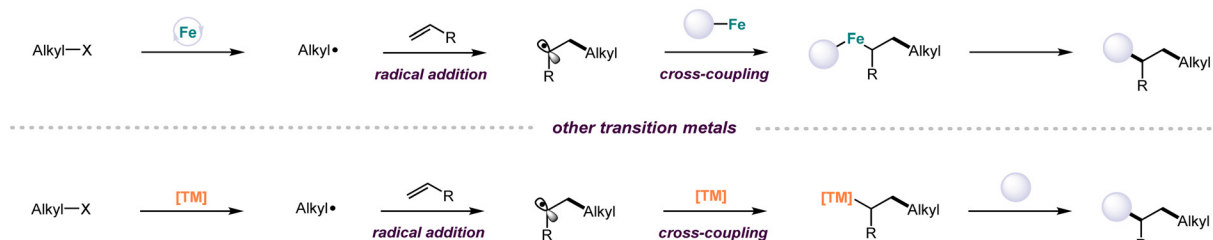
In the case of alkynes, current approaches mainly rely on *in situ*-generated alkenylmetal intermediates, which undergo subsequent coupling to introduce a second carbon substituent. Organomagnesium and organolithium reagents typically follow a migration-insertion pathway involving ate-complex formation. In contrast, organozinc reagents proceed *via* a unique ligand-directed mechanism, with regioselectivity governed by ligands such as NNN-pincer ligands, favoring addition to the less hindered carbon center of the alkyne. Notably, the regioselectivity in organomagnesium systems is highly dependent on heteroatom-directed strategies, enabling alkyl addition at positions distal to the heteroatom. However, the high reactivity of organolithium-derived ate complexes poses challenges in substrate scope and regioselectivity control.

Compared to alkene and alkyne catalytic systems, progress in the dicarbofunctionalization of allenes has been more limited, mainly due to the challenge of selectively addressing multiple reactive sites. Different organometallic reagents can influence the reaction pathway by modulating the oxidation state of the active iron species. Organozinc reagents often

generate iron(0) species, while organomagnesium reagents form ate-iron complexes that alter the electronic properties of the metal center. This organometal-dependent variation in oxidation state provides a new perspective for achieving selective control.

While iron shares certain reactivity patterns with other transition metals, accumulated evidence indicates that it also exhibits unique mechanistic profiles (Fig. 2). In the context of alkene difunctionalization, iron-catalyzed processes tend to proceed *via* radical addition followed by cross-coupling, whereas other metals, such as nickel, often operate through both radical addition/cross-coupling and carbometallation/cross-coupling pathways. For alkyne difunctionalization, iron catalysis typically favors a carbometallation/substitution sequence. In contrast, nickel and other metals more commonly follow carbometallation/cross-coupling pathways. This divergence can be attributed to the distinct behavior of the *in situ*-generated alkenyl-metal intermediates: in iron-catalyzed systems, these species are prone to undergo transmetalation with external organometallic reagents, generating new alkenyl-metal (*e.g.*, alkenyl-Zn) intermediates that subsequently participate in substitution-type reactions. By comparison, alkenyl-metal (*e.g.*, alkenyl-Ni) species derived from other transition metals more readily undergo oxidative addition with electrophiles to directly furnish the final coupling products. These mechanistic distinctions underscore the unique reactivity landscape of iron catalysis and suggest that iron may enable novel transformations that are less accessible with conventional transition metal systems. Moreover, due to differences in the activity of alkenyl-metal intermediates and the nature of the reaction pathways, iron catalysis is more likely to form carbon-heteroatom bonds (*e.g.*, C-I bonds) and to efficiently construct C(sp³)-C(sp³) bonds *via* substitution pathways. In contrast, other transition

a. Difunctionalization of unsaturated bonds via radical addition/cross-coupling



b. Difunctionalization of unsaturated bonds via carbometallation/substitution or carbometallation/cross-coupling

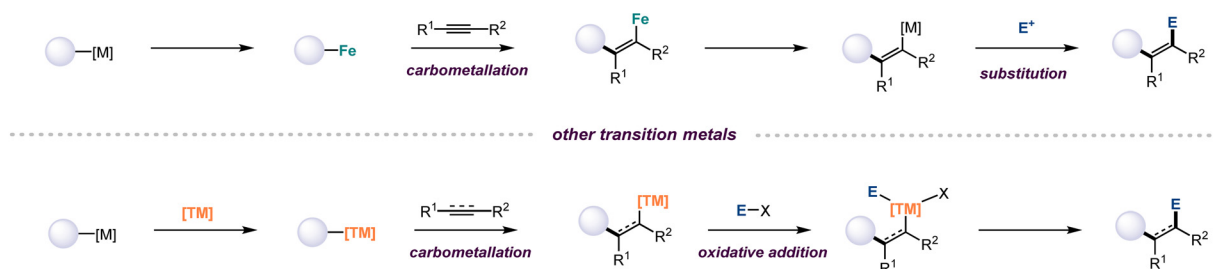


Fig. 2 Summary of difunctionalization of alkenes catalyzed by iron or other transition metals.

metals tend to favor the formation of $C(sp^3)-C(sp^2)$ bonds through oxidative addition processes.

Despite significant advancements in iron-catalyzed Negishi and Kumada-type dicarbofunctionalizations of alkenes, alkynes, and allenes, the development of three-component reductive couplings involving two electrophilic partners remains in its infancy. Future efforts in this area should focus on: (1) designing novel reductive coupling systems capable of overcoming the activation energy barriers associated with two electrophilic substrates; (2) advancing asymmetric iron-catalyzed dicarbofunctionalization by developing chiral ligands and modulating the electronic structure of the iron center to achieve stereocontrol; and (3) improving *in situ* mechanistic characterization by capturing crucial iron intermediates and combining experimental studies with theoretical calculations to elucidate structure–activity relationships tied to iron oxidation state changes and reaction selectivity. Furthermore, machine learning-assisted screening of ligand combinations may provide a promising paradigm for addressing current limitations in substrate scope and selectivity.

Conflicts of interest

There are no conflicts to declare.

Data availability

This review does not provide original data. All data can be found in the original literature.

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Notes and references

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